Dual-ligand fluorination

01 / Experiment

Use the two-page bench sheet. Every precise liquid transfer is robotic. Staff supply weighed chemicals/bulk liquids, handle caps and labeled labware moves, and import/start Python locally. Keep generated CSVs with their protocol.

Chemistry and design

Pyridine-2-sulfonyl fluoride → 2-fluoropyridine. Nominal per vial: 25 µmol SM, 1.5 µmol each ligand, 3 µmol Pd(COD)(DQ), 2.5 µmol mesitylene; 250 µL toluene, 95 °C, 500 rpm, 4 h under air. No base/external fluoride. Solvent addition defines nominal concentrations; solute displacement/contraction are accepted approximations.

R1: **48 pairs covering all 30 ligands**; L07 excluded. R2: **22 pairs +L17 tBuBrettPhos at 12 mol% total ligand**; L03 at most four wells. No repeats. Higher yield than the reference identifies a preliminary candidate; R1 comparisons also confound batch effects. Reproducibility and cooperativity require later matched controls.

Robot stock preparation

1. Generate `weighed_charges_template.csv`; enter **actual masses/assays**. Weigh ligands in the glovebox; preserve original bottles there. Pd goes directly into four 2 mL tubes: **4 A3/B3/C3, 22.8 mg each; 4 A4, 21.6 mg**. Total ≤90 mg; keep 10 mg for scope. On the OT-2 under air, the robot adds solvents/neat mesitylene, mixes and fills working sources. Stock preparation is inside the campaign.
2. **Live release is blocked:** bottom aspiration in the configured 15 mL tubes requires unverified nozzle entry; the conservative reach check fails. Obtain actual short, solvent-compatible stock containers and matching rack JSON, then regenerate and check reach/capacity. Follow carrier/raw-load CSVs; no invented glass-vial dimensions.
3. Run `prepare_stocks.py`: **p300 single left, p1000 single right**; tips 5 /8-9 respectively. Carriers occupy 3/6 temporarily; working racks 1/4. The supplied horizontal 8 ×30 mL reservoir in 2 is **single-channel only**. Robot prepares analytical matrix/masters/working stocks too.
4. Cap/store reserve carriers and the prepared analytical rack. Leave working tubes in mapped positions. Change right pipette to **p300 eight-channel GEN2**, replace tips 8/9 with 300 µL racks using the local App installation/calibration workflow. Keep left p300 single.

Slot	Reaction-stage use
1 / 4	2 mL working tubes; max 1.95 mL. Ligands at CSV addresses; neat IS **1 C2**, SM/IS **4 D2**, Pd **4 A3/B3/C3/A4**; prep acid **4 D3**. Retain RockPhos, SM/IS and all four Pd tubes.
2	New EMPTY 12-channel reservoir. Robot prepares workup fluids here during the hold; no staff premixing/refills.
3 / 10	Reaction block at pipetting / heater-shaker positions; manual moves. During holds, 3 temporarily takes raw solvents.
6	Calibration plate during R1; later a fresh final sample plate per round. Three compatible 96-well LC plates total.
5 / 8 / 9	Single / multi / multi tips. Keep 7 and 11 empty.

Confirm block/adapter/latch, offsets, <=45 mm cap clearance and liquid immersion with **1.5 mm bottom clearance**. Confirm LC plate/carrier/seal/needle depth and the requested full 1.7-min method/cycle, UV channel, RTs/LOQs. Remaining physical setup facts.

Run R1

Start R1 Python with **48 NEW EMPTY vials** in the block at 3. Robot doses **25 µL SM/IS +75 µL each ligand +75 µL Pd**. Retain SM/IS tube **4 D2**, capped between rounds: its initial **1.89 mL** comes from the same nominal 2 mL stock; no additional SM stock or refill. Expected remainder after R1: **689 µL**; R2 requires **at least 680 µL**. Record any loss.

Hand-cap all R1 vials; move/latch at App prompts. Run **95 °C /500 rpm /4 h**. During this hold the robot prepares **14 standards** and quench/diluent. Swap analytical rack into 4, raw reservoir into 3; seal/move calibration plate 6→HPLC. **Reaction stays on 10.** Restore rack 4 and remove raw reservoir when prompted.

Start a 4 h external timer at the App's hold-start message, in both rounds. Preparation and pauses count. Code checks/logs time but cannot stop heat while App-paused. At deadline, stop heat/shaking through the App; record deviations and do not resume dosing. The 60-second logging tolerance does not establish chemical equivalence.

Workup and final LC plates

1. Move sealed block to 3; **cool vials before opening**. App separately checks module <=30 °C. **Keep vials capped while robot pre-fills the fresh LC plate in 6 with 980 µL diluent/well.** Then uncap; robot adds **500 µL 9:1 ACN/water +0.15% FA**. Nominal extract: 750 µL, 0.1% acid, 1.25 mL headspace. Neither fluid contains IS.
2. **Hand-recap ALL vials**; shake **1500 rpm /2 min**, heater off. Return block to 3. Let bubbles settle; open the indicated vials for sampling.
3. Sample promptly after shaking; LC diluent is already in place. Robot uses a fresh tip, two slow same-well prewets, **21 µL aspiration at 1.5 mm**, 1 µL retained reverse surplus and 2 µL air gap. Dispense nominal **20 µL into prefilled LC wells**, noncontact; discard tip. No manual dilution/vial transfer.
4. Recap reaction vials. Move final LC plate **6→cooled shaker 10**, A1 orientation unchanged. Robot shakes **UNCOVERED, heater OFF, 1000 rpm /2 min** at the locally confirmed splash-free speed. Reaction block remains in 3.
5. Move LC plate **10→HPLC**, apply compatible pierceable seal and inject directly. R1: **R1-LC01 A1-H6**, 48 samples. R2: **R2-LC01 A1-H2 plus A3-G3**, 23 samples. **H3 is diluent only; never inject.** Follow generated sequence/maps; reaction and LC coordinates differ.

Final sample: 1 mL; nominal maximum analyte **666.7 µM**, mesitylene **66.7 µM**, FA **0.1%**, toluene **0.667%**. Use the calibrated injection volume. Seal after final shaking to limit evaporation. There are four final-sample plate moves plus the calibration-plate handoff.

R2 and finish

After ALL 48 R1 results pass QC, select 22 pairs +L17 reference. Run `refill_stocks.py` **before R2 dosing**. Remove block/LC plate from 3/6; present ORIGINAL bulk carriers by ID for robotic ligand top-ups. **No new weighing or hand aliquoting. Retain RockPhos and all four Pd tubes; no Pd refill.** Pd routing: R1 uses A3/B3/C3 for **18/18/12 doses**; R2 uses **six from C3, 17 from A4**. Exact solvent additions and each tube's capacity follow actual mass/assay.

Load **23 NEW EMPTY vials** in the SAME block; replace reservoir 2 with a NEW EMPTY one. Uncap required stocks. R2 Python doses **25 µL SM/IS each, then catalysts**; reference receives **150 µL L17**. Hand-cap and start the reaction. During R2 hold the robot makes fresh workup fluids; no new calibration. Repeat workup and final LC analysis.

Cap stock reserves between stages. Robot mixes sources before first use/every eight transfers, without air gaps/blowouts. Same-stock dosing tips may be reused only for validated noncontact transfers. Analytical samples always use fresh tips. Mixing can disperse solids; it does not prove a settling suspension is uniform.

Pause for visible layers, blocked tips, persistent bubbles, air uptake, dripping or tip contact. Do not sample an arbitrary layer, guess a top-up or filter/decant without recording the effect. Reprepare an uncertain analytical dilution robotically and record the replacement; retain rejected measurements. No separate homogeneity study is planned.

Finish after all 23 final injections and quantitative analysis. Save robot JSON logs, raw LC/MS/method/sequence files, actual weighed charges, times and deviations under `local_data/`. Run `python scripts/hte-return-package.py`; send its ZIP privately. It retains code, preparation/source ledgers, inputs, results, Claude records and file hashes; credentials are excluded.

02 / Stocks and calibration

Staff weigh; the OT-2 measures liquids. Enter actual masses/assays in the charges CSV; follow `bulk_carriers.csv`. Concentrations use nominal solvent addition; no hand aliquoting or volumetric top-up.

Reaction stocks

Stock	Preparation target at 100% assay	Per reaction
SM + mesitylene	**~322.32 mg SM /2 mL: 1 M SM +100 mM IS**. Robot adds ~27.82 µL neat IS +toluene; adjusted to actual mass/assays.	**25 µL**
Each ligand	**20 mM toluene**; weighed charges/reserves in CSV. Robot prepares bulk, mixes and fills working tubes.	**75 µL each**; reference 150 µL L17
Pd(COD)(DQ)	Direct in **4 A3/B3/C3: 22.8 mg each; A4: 21.6 mg**. Robot adds ~1.50/~1.43 mL toluene for **40 mM**. Total **90 mg**, ~5.94 mL; **10 mg for scope**.	75 µL
RockPhos L03	**~800 µL /20 mM**, 7.50 mg pure equivalent; 7.73 mg at 97%. Prepared directly in **1 C1**; retain that tube. **10 mg available, max 4 R2 wells.**	75 µL

Working tubes: 2 mL, max 1.95 mL. Load **1.89 mL SM/IS into 4 D2**; retain capped, no refill. Including losses: R1 uses **1201 µL**, leaves **689 µL**; R2 uses **576 µL**, leaves **113 µL** (R2 minimum: **680 µL**). Start with **48 empty R1 /23 new empty R2 vials**; dose 25 µL SM/IS before catalysts each round.

Acid: **4 D3**. Retain all four Pd tubes, without refilling: R1 **18/18/12** doses from A3/B3/C3; R2 **6 C3 +17 A4**. CSVs give exact additions from actual masses/assays, checked per tube and in total. Remaining 15 mL bulk sources fail conservative reach; supply suitable geometry. R2 ligand refills use original reserves; record losses.

Robot-made workup fluids

Each hold: robot makes 10% aqueous FA, then fluids in reservoir 2. Coarse-load raw solvents per CSV into A1-H1 reservoir in temporary slot 3.

Fluid	Composition	Use
Quench	**9:1 ACN/water, 0.15% v/v FA**, no IS	500 µL; capped 1500 rpm /2 min
LC diluent	**1:1 water/ACN, 0.1% v/v FA**, no IS	980 µL +20 µL extract

Robot-made authentic standards

Weigh separate FIT and CHECK charges per analyte: **at least 5 mg**, assay-corrected; enter actual masses. Weigh liquid product by difference. Robot makes **10 mM** masters (SM ~6.45 mg /4 mL; product ~5.83 mg /6 mL), matched matrix and **0.5/5 mM FIT dilutions**. Retain reserves.

Robot makes **20 mM mesitylene from 25 µL neat IS**, then **2 mM** working stock. Density: 0.864 g/mL at 25 °C; configure assay.

Calibration during hold one - 14 injections

Robot fills/pipette-mixes **CALIBRATION**: SM A1-G1, product A2-G2. Seal/move directly to LC during R1. Follow generated preparation and injection maps.

Analyte, µM	Role	Stock, mM	Analyte, µL	Matrix, µL
0	Fit blank	None	0	966.667
10	Fit	0.5	20	946.667
50	Fit	0.5	100	866.667
150	Fit	5	30	936.667
350	Fit	5	70	896.667
800	Fit	5	160	806.667
666.667	Independent check	10, CHECK	66.667	900.000

Add **33.333 µL IS to every row**. Fit blank +five levels, exclude CHECK. Require **R^2 >=0.995**, **check error <=15%**, fit recovery within **15%** (**20%** at 10 µM), measured separation/LOQs. Quantitative floor: greater of measured LOQ and lowest nonzero calibrator. Local Python converts integrated UV areas to yields. LC processing.

03 / Local pipeline

Frozen **48 pairs**: `campaigns/hackathon2026/round1_design.csv`. Python quantifies UV data; cached Claude scoring selects R2. Stock preparation uses no API calls. Physical actions: bench sheet.

Install and configure

Python 3.12: `uv sync --python 3.12 --extra sources --extra robot`; prefix commands with `uv run`. Or install `.[sources,robot]` in a venv. Copy config to `local_data/campaign.json`, calibration template to `local_data/calibration-plan.csv`, key example to `anthropic.key`. Put the raw Claude key there; preserve existing keys. Never send it to the robot/Git.

Use schema **5**; replace old protocols while preserving measured LC facts. Configure method/channel, full cycle, RTs/LOQs and physical evidence. **Default bulk-source reach fails:** obtain actual suitable short-container/rack geometry before live release. Concentrations use nominal solvent addition. Twelve hours is advisory; record real times.

Robot preparation, then R1

Generate a guarded preview; fill actual masses/assays in `local_data/charges.csv` from its template. Pd: **22.8 mg each in 4 A3/B3/C3; 21.6 mg in 4 A4**, 90 mg total. Follow carrier/raw-load CSVs. After local checks:

```
python -m hte.cli --config local_data/campaign.json prepare-stocks \
  --output output/prep-preview
python -m hte.cli --config local_data/campaign.json prepare-stocks \
  --charges local_data/charges.csv --output local_data/stock-prep --live
```

Run `local_data/stock-prep/prepare_stocks.py` locally: p300 single left /p1000 single right. Preserve its manifest at the default configured path. Follow swaps; afterward change right to p300 multi and tips 8/9 to 300 µL through the App installation/calibration workflow.

```
python -m hte.cli --config local_data/campaign.json export \
  --design campaigns/hackathon2026/round1_design.csv \
  --output output/r1-dose --live --stage dose_and_react
```

Start with **48 NEW EMPTY R1 vials**. Robot adds 25 µL SM/IS per vial, then catalysts, and heats the capped block. Retain **SM/IS tube 4 D2**, capped between rounds; its 1.89 mL initial fill covers both rounds. Robot makes standards/workup fluids during the hold. Send calibration plate to LC using generated map. Import/calibrate; set `lc_calibration_validated=true` after success. Export R1 workup with the SAME command/design, changing to `--output output/r1-workup --stage workup`. Keep prepared reservoir 2; use a fresh final LC plate.

R2 and finish

Complete actual injection maps. After ALL 48 results:

```
python scripts/hte-lc-results.py next \
  --dosing output/r1-dose/dosing.csv \
  --exports local_data/r1-batch*.csv \
  --injections local_data/r1-map*.csv --output output/r1-decision
```

Calibration defaults to `local_data/calibration.csv`. QC precedes Claude: **387 untested pairs →22 +L17 reference**, L03 max four. **After interruption rerun the SAME command/output**; accepted scores are reused. Inspect uncertain API requests; a new folder may repeat charges. Run printed `refill_stocks.py` **FIRST** for ligand top-ups; retain SM/IS **4 D2** and all four Pd tubes without refilling. Fit **23 NEW EMPTY vials** in the same block and a NEW EMPTY reservoir 2; R2 Python adds **25 µL SM/IS per vial, then catalysts**. If unreleased, export `next_design.csv` with R1 `--history`, `--live` and a new export folder. After 23 final injections, run helper `final`; no Claude.

Return records

Keep actual weights, preparation/source ledgers, LC exports/maps, calibration/QC, Claude records and matching doses/protocols. Copy robot JSON to `local_data/robot_logs/`, raw LC/MS/method files to `local_data/instrument_raw/`, times/deviations to `local_data/run-notes.txt`. Stop writers; `python scripts/hte-return-package.py` creates `transfer/hte-results.zip` with hashes, credentials excluded. Send privately; Git does not preserve local run data.